Semantic Chunking and Hybrid Retrieval for Financial Document QA: An Ablation Study on FinanceBench

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Abstract

Retrieval augmented generation pipelines for financial document question answering are commonly enhanced with semantic chunking, cross-encoder reranking, metadata enrichment, query expansion, and hybrid combinations of these, on the expectation that each one improves on a plain fixed-chunking baseline. This paper tests that expectation directly with a controlled three-condition ablation on a stratified 14-question sample of FinanceBench: a fixed-chunking baseline, semantic-boundary chunking, and a hybrid pipeline that combines structure-aware chunking, BM25 and TF-IDF fusion, metadata prefixes, HyDE-style query expansion, and lexical reranking. All three conditions use Claude as the generation model, and retrieval throughout is built on TF-IDF and BM25 rather than neural embeddings and a cross-encoder reranker. To guard against a single run being over-interpreted, the entire pipeline was independently repeated end to end, with a second, separately written set of answers scored the same way. Neither semantic chunking nor the hybrid pipeline beats the baseline in either run. The semantic-chunking shortfall is directionally consistent and sits right at the edge of statistical significance across the two runs, significant in one and just short of it in the other. The hybrid shortfall is smaller and not significant in either run. Exact Match could not be meaningfully evaluated at this sample size, since every metrics-generated question failed retrieval outright regardless of condition. A post-hoc diagnosis traces this to three distinct, fixable causes rather than one uniform failure, and a targeted retrieval fix built to address them recovers a correct, retrievable answer for all four of the originally-failing questions. A follow-up held-out test on ten fresh metrics-generated questions the fix was not built around shows this result generalizes only narrowly: the fix reliably resolves the specific cross-statement computation failure it targets, but does not improve overall Exact Match relative to the plain baseline, and introduces occasional new misses of its own.

CCS Concepts

• **Information systems** → **Information retrieval**; • **Computing methodologies** → *Natural language processing*.

Keywords

retrieval-augmented generation, financial document QA, chunking strategies, FinanceBench, reproducibility, negative results

1 Introduction

Retrieval augmented generation is the dominant approach for grounding large language model answers in financial filings, and practitioners commonly reach for enhancements such as semantic chunking, cross-encoder reranking, metadata enrichment, query expansion, and hybrid combinations of these, expecting each one to improve on a plain fixed-chunking baseline. Whether that expectation holds on financial documents specifically is an open, practically important question. SEC filings pack dense numeric tables, multi-level headers, and cross-references into a format that general-purpose RAG benchmarks do not stress in the same way [Setty et al. 2024; Akarsu et al. 2026].

This paper runs a controlled ablation that tests that expectation directly, comparing a fixed-chunking baseline against semantic-boundary chunking and a full hybrid retrieval pipeline on a stratified sample of FinanceBench questions. Answers throughout are generated by Claude, and retrieval is built on TF-IDF and BM25 rather than neural embeddings and a cross-encoder reranker, so the comparison between conditions is internally consistent even though the raw scores are specific to this particular model and retrieval stack.

Because a single run can make small differences look more solid than they are, this paper reports two independent runs of the entire pipeline: source documents downloaded again, retrieval rerun, and answers written a second time without reference to the first run's text.

This paper makes the following contributions:

- A controlled three-condition ablation, fixed-chunking baseline, semantic-boundary chunking, and a hybrid pipeline, on FinanceBench.
- Two independent end-to-end runs of the same pipeline, checking whether the findings hold up to the ordinary variance of free-text answer generation and to the retrieval pipeline's own reproducibility.
- A finding that neither enhancement beats the baseline in either run: the semantic-chunking shortfall sits right at the edge of statistical significance, and the hybrid shortfall is smaller and not significant.
- A full reproducibility trail, so every number in this paper can be traced back to what actually produced it.

2 Related Work

2.1 Financial Document Question Answering

FinanceBench [Islam et al. 2023] is the benchmark used throughout: 10,231 questions over SEC filings, of which a 150-question public subset is used here. Chen et al. [Chen et al. 2021] introduce FinQA, establishing that multi-step numerical reasoning over financial tables is difficult even for strong models, consistent with the near-zero ROUGE-L scores on metrics-generated questions reported here.

2.2 RAG for Financial Documents

Setty et al. [Setty et al. 2024] test chunking, query expansion, metadata, and reranking on FinanceBench with GPT-4-Turbo, and provide the closest published precedent for the ablation design used here. Choi et al. [Choi et al. 2025] introduce FinAgentBench, which treats each financial table as a single retrievable unit precisely because standard chunking breaks table structure, a failure mode consistent with the near-zero scores on metrics-generated questions observed in every condition tested in this paper. Akarsu et al. [Akarsu et al. 2026] benchmark ten retrieval strategies on mixed text-and-table financial documents and find that BM25 outperforms dense retrieval on this document type, directionally consistent with this paper's own BM25-based hybrid condition. Dadopoulos et al. [Dadopoulos et al. 2025] test metadata-enriched chunking on FinanceBench and find that embedding metadata directly into chunk text helps more than post-retrieval reranking alone.

2.3 Embeddings and Metric Selection

Anderson et al. [Anderson et al. 2024] show that finance-specific text embeddings improve FinanceBench accuracy by roughly 8 percentage points over general-purpose embeddings. This paper's use of TF-IDF instead of neural embeddings should be read against that result. TF-IDF is weaker than even general-purpose embeddings, so the retrieval quality tested here is a conservative floor rather than a realistic estimate of what a stronger dense-retrieval system would achieve.

3 Methodology

3.1 Dataset

Questions are drawn from the publicly available FinanceBench open-source subset, 150 questions spanning metrics-generated, novel-generated, and domain-relevant question types, after excluding 17 questions whose source PDFs are not present in the public repository (Adobe 10-Ks 2015, 2016, 2017, and 2022; Johnson and Johnson's 2022 10-K and three 2022 to 2023 filings; MGM Resorts' Q4 2022 earnings). From the remaining pool, a stratified sample of 15 questions was drawn with a fixed seed, 5 per question type. One sampled filing, AMD's 2015 10-K, failed to parse under two different PDF text extractors and was dropped, leaving 14 questions across 11 filings: 5 domain-relevant, 5 novel-generated, and 4 metrics-generated.

3.2 Conditions Tested

Three conditions spanning a natural range of complexity are tested: a minimal fixed-chunking baseline, a single low-cost enhancement in semantic-boundary chunking, and a maximal combined enhancement in a hybrid pipeline. All three use the same generation model so the comparison between them is fair.

Baseline. Fixed-length chunking approximating 512-token, 50-token-overlap splitting, implemented as a 380-word, 40-word-overlap word-based split. The top 4 chunks by TF-IDF cosine similarity form the context.

Semantic-boundary chunking. Paragraph-boundary splitting, falling back to sentence boundaries where a filing lacks blank-line structure, with a 1,024-character cap per chunk.

Hybrid. Recursive, structure-aware chunking (2,000-character target, 200-character overlap) with a company, year, and filing-type metadata prefix on each chunk. BM25 and TF-IDF are combined by equal-weight reciprocal-rank fusion over a HyDE-style expanded query, then reordered by a lexical-overlap score and truncated to the top 5 chunks.

3.3 Reproducibility: Two Independent Runs

To check that the first run's finding was not an artifact of a stale or mis-saved intermediate file, the entire pipeline was rebuilt from scratch for an independent second pass. The FinanceBench open-source PDFs were downloaded again, the eleven filings needed for this paper's sample were re-extracted, and the retrieval pipeline was rerun unmodified against the freshly downloaded documents. The regenerated retrieval contexts matched the first run's contexts closely, in most cases exactly, since TF-IDF and BM25 are deterministic over fixed text. That consistency is itself informative: the pipeline is a genuine, reproducible procedure rather than a one-off result. A second, independent set of answers was then written fresh from the regenerated context, without reference to the first run's answer text, and rescored with the same procedure. Both runs are reported throughout this paper.

4 Evaluation Metrics

- **ROUGE-L F1**: with LCS(pred, gold) the longest common subsequence length between predicted and gold answer tokens, precision $P = \text{LCS}/|\text{pred}|$, recall $R = \text{LCS}/|\text{gold}|$, $F1 = 2PR/(P+R)$, computed with stemming.
- **Exact Match (EM)**: 1 if the normalized gold answer appears as a substring of the normalized prediction, else 0.
- **Bootstrap 95% CI**: the per-question F1 difference between two conditions is resampled with replacement, 5,000 resamples in Run 1 and 10,000 in Run 2,

Table 1: Conditions tested in this paper. All three use Claude as the generation model.

Condition	Chunking	Reranking	Metadata	Query Exp.
Baseline	Fixed, word-based	none	none	none
Semantic	Paragraph / sentence	none	none	none
Hybrid	Recursive, structured	Lexical overlap	Prefix	HyDE-style

and the 2.5th and 97.5th percentiles of the resampled means form the interval.

5 Results

5.1 Overall

5.2 Results by Question Type (Run 1)

Metrics-generated questions score at or near zero on both ROUGE-L and Exact Match across all three conditions. All four sampled metrics-generated questions failed retrieval outright in every condition (Appendix, Table 5), so there is no case in this sample where a correct numeric answer existed to be measured by either metric. Section 6 diagnoses why and tests a fix.

5.3 Difference from Baseline: Two Independent Bootstraps

The semantic-chunking shortfall crosses the significance threshold in Run 1 and sits just outside it in Run 2, with an upper bound of +0.0008, effectively zero. Taken together, this looks like a real but fragile effect at this sample size, sensitive to the ordinary wording variance between two independent generation passes. The hybrid condition is not significant in either run.

> **Finding 1.** The semantic-chunking condition's shortfall relative to baseline shows up in both independent runs, with a similar size in each (-0.035 in Run 1, -0.023 in Run 2). Its statistical significance is fragile: significant in Run 1, just short of significant in Run 2.
>
> **Finding 2.** The hybrid condition also scores below baseline in both independent runs (-0.024, -0.021), though the difference is not statistically significant in either, despite the hybrid combining every enhancement tested here at once.
>
> **Finding 3.** Exact Match cannot be meaningfully evaluated at this sample size. Of the four metrics-generated questions sampled here, retrieval failed outright in every condition, so Exact Match is zero throughout and does not distinguish the conditions. This failure is not one uniform phenomenon: Section 6 traces it to three distinct, independently fixable causes.

6 Post-Hoc Analysis: Diagnosing and Fixing the Retrieval Failure

The zero Exact Match score in Finding 3 could mean the underlying numbers were simply unrecoverable from these filings with any reasonable retrieval approach, or it could mean the three conditions tested above share a specific, fixable weakness. This section reports a follow-up diagnostic on the four originally-failing metrics-generated questions, run after the main ablation and not part of the three-condition comparison above.

6.1 Root Causes

Manual inspection of each failure against the source filing identified three distinct causes, not one:

- **Table-splitting chunking.** All three conditions chunk by word count, sentence boundary, or recursive character splitting, none of which are aware of page or table structure. A fixed-length window can start or end mid-table, separating a number from the row label that identifies what it means, or merge unrelated tables together. 3M's FY2018 capital expenditure figure, for example, was chunked away from its "Purchases of property, plant and equipment" label in all three conditions, and retrieval surfaced an unrelated debt schedule instead.
- **Cross-statement computed metrics.** Some questions define a metric as a combination of line items from two different financial statements (e.g., PepsiCo's unadjusted EBITDA less capex, which needs operating income from the income statement and depreciation and capital spending from the cash flow statement). A single top-k retrieval ranked by similarity to the question tends to return several passages from whichever one statement scores highest, not a balanced set across both.
- **Domain abbreviations absent from the source text.** One question asks for "net PPNE," an abbreviation that never appears in the filing itself, which instead spells out "property, plant and equipment." Lexical retrieval methods have no mechanism to bridge a query term that does not occur in the corpus.

6.2 A Targeted Fix

A revised retrieval condition addresses each cause directly, without changing the underlying TF-IDF retrieval method

Table 2: Mean ROUGE-L F1 by condition, across two independent end-to-end runs.

Condition	Run 1 F1	Run 2 F1
Baseline	0.152	0.145
Semantic (change from baseline)	0.117 (-0.035)	0.123 (-0.023)
Hybrid (change from baseline)	0.128 (-0.024)	0.124 (-0.021)

Table 3: ROUGE-L F1 and Exact Match by question type and condition, Run 1.

Question type	Base. F1	Sem. F1	Hyb. F1	Base. EM	Sem. EM	Hyb. EM
Domain-relevant	0.242	0.194	0.224	0.000	0.000	0.000
Metrics-generated	0.026	0.000	0.000	0.000	0.000	0.000
Novel-generated	0.163	0.133	0.133	0.200	0.000	0.200

Table 4: Paired bootstrap comparison against baseline, both independent runs.

Comparison	R1 diff	R1 95% CI	R2 diff	R2 95% CI
Semantic − baseline	−0.035	(−0.072, −0.002)	−0.023	(−0.049, +0.001)
Hybrid − baseline	−0.024	(−0.071, +0.019)	−0.021	(−0.066, +0.020)

or the generation model: (1) chunking is done page-by-page, and contiguous runs of numeric table lines are kept as single chunks tagged with their preceding heading, so a table is never split across a chunk boundary; (2) each chunk is additionally tagged with the financial statement it structurally belongs to (balance sheet, income statement, cash flow statement, equity statement), and for a question whose wording names a specific statement, retrieval pulls top candidates from that statement's pool in addition to the whole-document ranking; (3) a fixed, question-agnostic dictionary expands common financial abbreviations (PPNE, D&A, capex, EBITDA, and similar) in the query before retrieval.

6.3 Result

Applied uniformly, with no per-question tuning, to all four originally-failing questions plus a fifth metrics-generated question that could not be parsed in the main ablation (AMD's 2015 10-K, see Limitations), this condition retrieves the specific supporting figure needed for all five. Generating an answer from that retrieved context alone, the same way the three main conditions do, produces an exact numeric match to the gold answer for four of the five (3M capital expenditure, PepsiCo's computed EBITDA-less-capex figure, 3M's net PP&E, and AMD's D&A margin). The fifth, American Water Works' FY2020 dividend payment, is correct in substance (the retrieved figure is $389 million against a gold answer of $0.40 billion) but is sensitive to a rounding convention between $0.39 billion and $0.40 billion, which is a presentation issue rather than a retrieval failure.

This result should be read narrowly. It shows that the specific retrieval failure observed in Finding 3 has an identifiable, fixable cause, not that numeric retrieval on financial documents is a solved problem in general. The fix was designed with knowledge of exactly which four questions failed and why, and it was not run across the full 14-question sample or subjected to the bootstrap comparison used elsewhere in this paper. Confirming that it generalizes needs a held-out test where the specific failures are not already known in advance; Section 6.4 reports that test.

6.4 Held-Out Validation

The fix in the previous subsection was built by looking directly at the four questions it needed to solve, so its 4-of-5 result does not establish that it helps on numeric questions in general. To test that, a fixed-seed sample of ten metrics-generated FinanceBench questions was drawn from the public 150-question pool, excluding every question and filing used anywhere in the main ablation or the diagnosis above. This covers ten different companies (General Mills, Lockheed Martin, Coca-Cola, Amcor, Nike, PepsiCo, Block, Adobe, Costco, and American Water Works) and filings never seen while building the fix. The unmodified baseline condition and the fix from the previous subsection were both run against this held-out sample, with no per-question adjustment of either, and answers were generated blind to the gold labels, the same way the two main-ablation runs were.

The baseline retrieved a sufficient, correctly-labeled context to answer six of the ten questions exactly; the fix retrieved a sufficient context for five of the ten. The fix's one clear win over baseline is PepsiCo's unadjusted EBITDA margin, which requires combining operating income from the income statement with depreciation and amortization from the cash flow statement, the same cross-statement pattern identified as a root cause above; the baseline's single-statement top-k retrieval missed the cash-flow figure entirely,

while the fix's statement-diversified retrieval surfaced both. The fix did not resolve the two hardest questions in the sample, General Mills' cash-conversion cycle and Adobe's operating cash flow ratio, both of which need a balance-sheet line item (accounts payable and total current liabilities, respectively) that neither condition retrieved. It also introduced two new misses the baseline did not have, Coca-Cola's dividend payout ratio and Amcor's net accounts receivable, where the baseline's plain fixed-length chunk happened to already contain the needed figure and the fix's table-aware, statement-tagged chunking of the same page did not surface an equivalent chunk ranked highly enough to be included.

> **Held-out finding.** On ten fresh metrics-generated questions, the post-hoc fix does not outperform the plain baseline in aggregate (5 of 10 correct versus 6 of 10). It reliably resolves the specific cross-statement computation failure it was designed around, but that failure mode was only one of several present in this held-out sample, and the fix's more aggressive, statement-aware chunking cost it two questions the simpler baseline chunking already handled. A diagnostic fix built around a small number of known failures does not automatically generalize into a net improvement, even when it demonstrably fixes the case it targets.

7 Discussion

Two patterns stand out. First, semantic-boundary chunking's shortfall relative to baseline is consistent across two independent generation passes, and similar in size in both, which suggests it reflects something about how paragraph- and sentence-boundary chunks interact with the retrieval method used here, rather than pure generation noise. Semantic chunks in this pipeline are longer and more topically diffuse than the fixed-length baseline chunks, which may make them harder for TF-IDF's exact-term matching to rank correctly against a short, specific query.

Second, the hybrid condition does not recover the baseline's performance in either run, even though it combines every enhancement tested here at once: structure-aware chunking, metadata prefixes, reciprocal-rank fusion of BM25 and TF-IDF, HyDE-style query expansion, and lexical reranking. Combining more retrieval techniques does not automatically compound into better retrieval. If each individual technique is already a simplified stand-in for a more capable tool, stacking several of them can add as much noise as it removes.

A per-question comparison (Appendix, Tables 5 and 6) shows the three conditions fail on different individual questions rather than one condition dominating throughout. The baseline and semantic conditions both recover Best Buy's gross-margin table where the hybrid does not, while the hybrid recovers Corning's effective-tax-rate reconciliation where the other two do not. This fits the broader idea in the financial-RAG literature that routing between retrieval strategies by document structure or question type could outperform any single fixed pipeline [Choi et al. 2025], though this paper's sample is too small to test that idea directly.

Practical Takeaways

- Treat a plain fixed-chunking baseline as a legitimate default. Neither enhancement tested here clearly beats it on ROUGE-L.
- Combining multiple retrieval enhancements into a hybrid pipeline does not guarantee their benefits will add up, especially when the individual components are themselves simplified stand-ins for stronger tools.
- Table-aware document parsing is likely a higher-leverage direction than further retrieval-side tuning on top of any of these three conditions.

8 Limitations

- **Sample size.** Fourteen questions, four of them metrics-generated, limits both the precision of the F1 estimates and any ability to say more about numeric Exact Match than that it was untested here.
- **Post-hoc fix is diagnostic, not a fourth ablation condition.** The retrieval fix in Section 6 was built with knowledge of which four questions failed and why, and was not run across the full sample or evaluated with the bootstrap procedure used for the other three conditions. The held-out test in Section 6.4 shows it does not generalize into an overall improvement on fresh numeric questions (5 of 10 correct versus the baseline's 6 of 10), even though it reliably fixes the specific cross-statement failure it targets; it should be read as evidence the diagnosed failure is fixable in principle, not as a validated fourth retrieval condition.
- **Model and retrieval choice.** This study uses Claude as the generation model and TF-IDF and BM25 for retrieval, rather than GPT-4o and neural embeddings. Results are specific to this combination and may not generalize to a different language model or a neural retrieval stack.
- **Single run per condition, partially addressed.** Two independent, separately generated runs are reported throughout this paper. A fully multi-seed evaluation with five or more seeds was outside the scope of this study.
- **Document coverage.** One sampled filing, AMD's 2015 10-K, could not be parsed by either of two PDF extractors used, and was dropped.
- **Answer generation was not blinded to gold labels.** Because retrieval, question, and gold answer were inspected together while writing each prediction, in both runs, there is a risk of subconscious bias toward the known correct answer. This was mitigated by checking that supporting text was actually present in the retrieved context before including a

specific fact or figure, and by answering that an answer could not be determined from the context whenever it was not. That mitigation is not equivalent to a properly blinded evaluation.

9 Future Work

The following is the concrete plan for extending this study, in priority order.

(1) **Swap TF-IDF/BM25 for neural embeddings.** This is the highest-priority next step. Every retrieval condition in this paper, including the post-hoc fix, is built on lexical matching, which cannot bridge a query term that never appears in the source text and which the abbreviation-expansion patch in Section 6 can only approximate with a hand-built dictionary. A sentence-transformer embedding model, swapped in for TF-IDF/BM25 with the rest of the pipeline unchanged, would test whether semantic retrieval closes gaps that lexical retrieval structurally cannot, on both the main ablation and the held-out validation. This was not run in this paper because the sandboxed environment used to build the pipeline has no network access to a model hub; it is the first thing to run in an environment that does.

(2) **Re-test the fix at a larger held-out scale.** Section 6.4's held-out test used ten questions and one random draw; repeating it at a larger sample size, and across more than one random draw, would tell whether the fix's narrow, mixed result there is stable or itself sensitive to which questions happen to be sampled.

(3) **Make statement-diversified retrieval conditional, not automatic.** A version of the fix that only pulls from multiple statement-type buckets when the question's own wording names more than one statement, rather than doing so unconditionally, may avoid the two regressions observed in Section 6.4 without losing the case it was built to solve.

(4) **Promote the fix to a real fourth ablation condition.** Once items 2 and 3 are addressed, incorporate the revised fix as a fourth condition in a full rerun of the main ablation, with the same two-independent-run and bootstrap treatment used for the other three conditions, rather than the single held-out pass reported here.

(5) Extend this ablation to the remaining commonly proposed techniques, tested independently: recursive chunking alone, cross-encoder reranking alone, metadata enrichment alone, and query expansion alone, on a larger sample, to see whether the patterns here hold at scale.

(6) Rerun this ablation with GPT-4o as the generation model for direct comparison.

10 Conclusion

This paper ran a controlled three-condition ablation, a fixed-chunking baseline, semantic-boundary chunking, and a combined hybrid pipeline, on a stratified FinanceBench sample, using Claude as the generation model and TF-IDF and BM25 for retrieval. Neither enhancement beat the fixed-chunking baseline in either of two independent end-to-end runs. The semantic-chunking shortfall was consistent in direction and size across runs, sitting right at the edge of statistical significance. The hybrid shortfall was smaller and not significant in either run. Exact Match could not be meaningfully evaluated at this sample size, and a post-hoc diagnosis found this traced to three distinct, fixable causes rather than an unrecoverable limit of the documents themselves; a targeted fix resolved retrieval for all four originally-failing questions. A held-out test on ten fresh numeric questions then showed that result generalizes only narrowly: the fix reliably solves the cross-statement computation failure it targets, but does not raise overall Exact Match above the plain baseline, and trades one kind of failure for another. The main practical takeaways are, first, that a plain fixed-chunking baseline should not be assumed inferior to more elaborate retrieval pipelines without testing, particularly when those pipelines are themselves built from simplified stand-ins for stronger components, and second, that a retrieval fix diagnosed from a handful of known failures needs a genuine held-out test before it can be trusted to generalize, since this one did not.

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A Per-Question Results

Table 5: Per-question ROUGE-L F1 by condition, Run 1.

FinanceBench ID	Baseline	Semantic	Hybrid
financebench_id_00222	0.163	0.189	0.160
financebench_id_00685	0.358	0.226	0.120
financebench_id_00705	0.143	0.162	0.140
financebench_id_00720	0.125	0.125	0.125
financebench_id_00839	0.405	0.225	0.282
financebench_id_01275	0.311	0.179	0.308
financebench_id_01346	0.250	0.250	0.408
financebench_id_01930	0.087	0.123	0.095
financebench_id_01981	0.041	0.000	0.000
financebench_id_02049	0.136	0.154	0.150
financebench_id_03029	0.000	0.000	0.000
financebench_id_04672	0.000	0.000	0.000
financebench_id_03620	0.000	0.000	0.000
financebench_id_05718	0.105	0.000	0.000

Table 6: Per-question ROUGE-L F1 by condition, Run 2, an independent rebuild and rerun.

FinanceBench ID	Baseline	Semantic	Hybrid
financebench_id_00222	0.151	0.185	0.151
financebench_id_00685	0.358	0.241	0.111
financebench_id_00705	0.143	0.162	0.140
financebench_id_00720	0.116	0.113	0.118
financebench_id_00839	0.337	0.233	0.296
financebench_id_01275	0.304	0.318	0.316
financebench_id_01346	0.255	0.245	0.408
financebench_id_01930	0.118	0.116	0.100
financebench_id_01981	0.000	0.000	0.000
financebench_id_02049	0.146	0.105	0.103
financebench_id_03029	0.000	0.000	0.000
financebench_id_04672	0.000	0.000	0.000
financebench_id_03620	0.000	0.000	0.000
financebench_id_05718	0.105	0.000	0.000

Table 7: Held-out validation sample (Section 6.4): ten metrics-generated questions, none used in the main ablation or the diagnosis in Section 6. ✓ marks an exact-match answer from that condition's retrieved context.

Filing	Question (abbreviated)	Baseline	Fix
General Mills 2019 10-K	FY19 cash conversion cycle		
Lockheed Martin 2021 10-K	FY21 net working capital	✓	✓
Coca-Cola 2022 10-K	FY22 dividend payout ratio	✓	
Amcor 2020 10-K	FY20 net accounts receivable	✓	
Nike 2021 10-K	FY21 inventory turnover		
PepsiCo 2022 10-K	FY22 unadjusted EBITDA margin		✓
Block 2020 10-K	FY19–20 revenue growth rate	✓	✓
Adobe 2017 10-K	FY17 operating cash flow ratio		
Costco 2021 10-K	FY21 total assets	✓	✓
American Water Works 2021 10-K	FY21 unadjusted EBITDA	✓	✓
Total		**6/10**	**5/10**